

PROPOSED PILOT • SYNTHETIC PROTOTYPE

CyberPulse AI

Anticipate the next cash-out.

Predictive location and time-window forecasting for cyber-fraud investigations.

THE PILOT QUESTION

Can ranked cash-out areas and time windows help investigators prioritise leads faster than historical-hotspot methods?

WHERE

Candidate geographic areas

WHEN

2–4-hour estimated windows

WHY

Model feature explanations

Current model scope: H3 resolution 8 areas; a 24-hour horizon from the complaint timestamp. Forecasts are leads for review, not confirmed withdrawal addresses.

REQUESTED NEXT STEP

A technical review with RBIH and a scoped, authorised data evaluation with suitable institutional partners.

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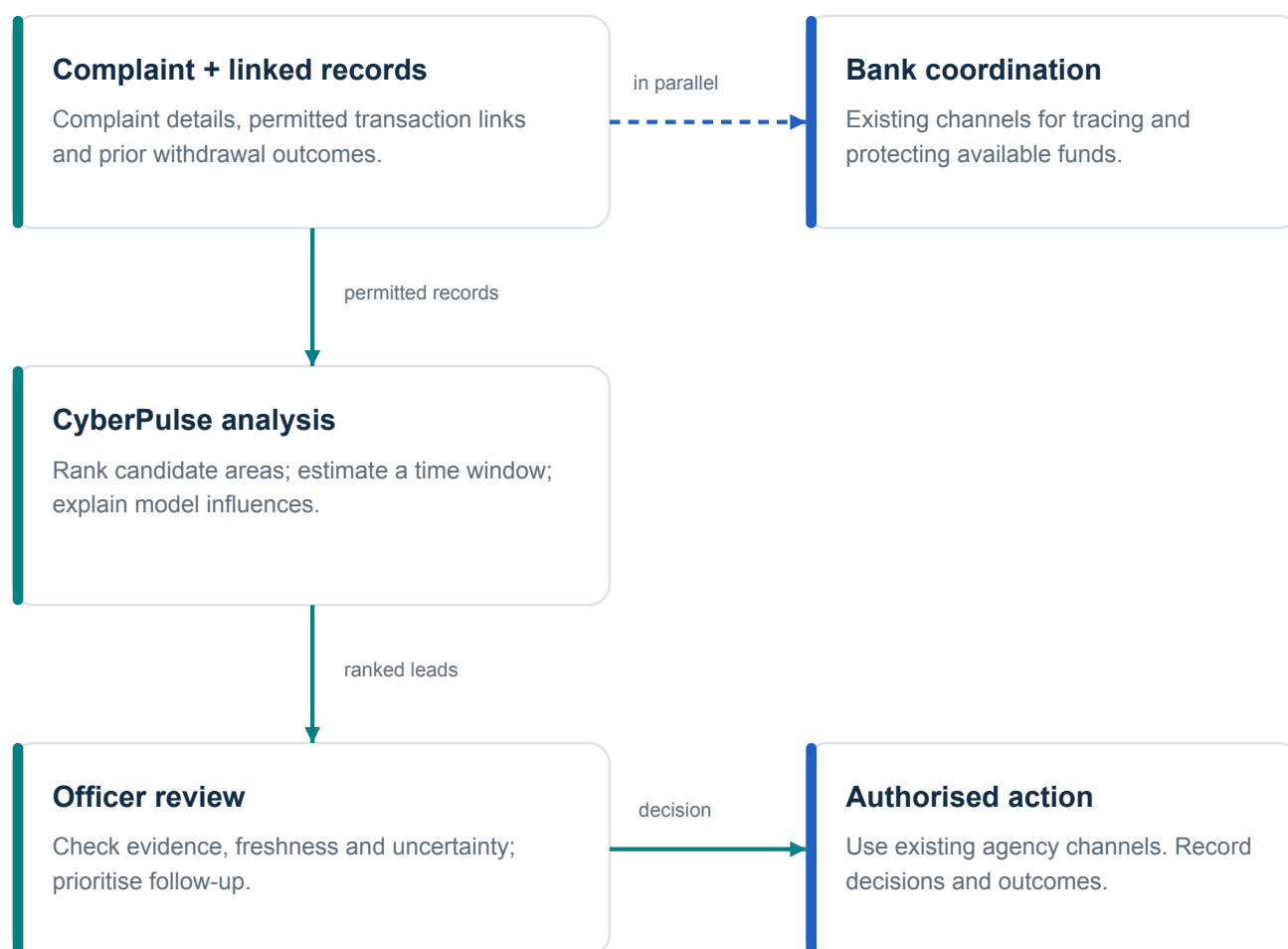
18 September 2026 • Problem Statement 26184 • Revision 2

01 / ECOSYSTEM ALIGNMENT

Where CyberPulse fits

A proposed decision-support step using authorised case records. Existing bank coordination and fund-protection efforts continue in parallel. [1, 2]

PROPOSED WORKFLOW • NO LIVE INTEGRATION



PROPOSED CONTRIBUTION

Help an investigator choose which lead to examine next. An area forecast does not identify an accused person or justify sending police to every ATM.

RBIH's MuleHunter.ai focuses on mule-account detection. The May 2026 I4C–RBIH MoU supports intelligence sharing. CyberPulse proposes complementary cash-out forecasting; access to those systems is not assumed. [2, 3]

From forecast to review

Illustrative interface composition based on the current map workflow. All scores, locations and ATM markers below are synthetic examples.

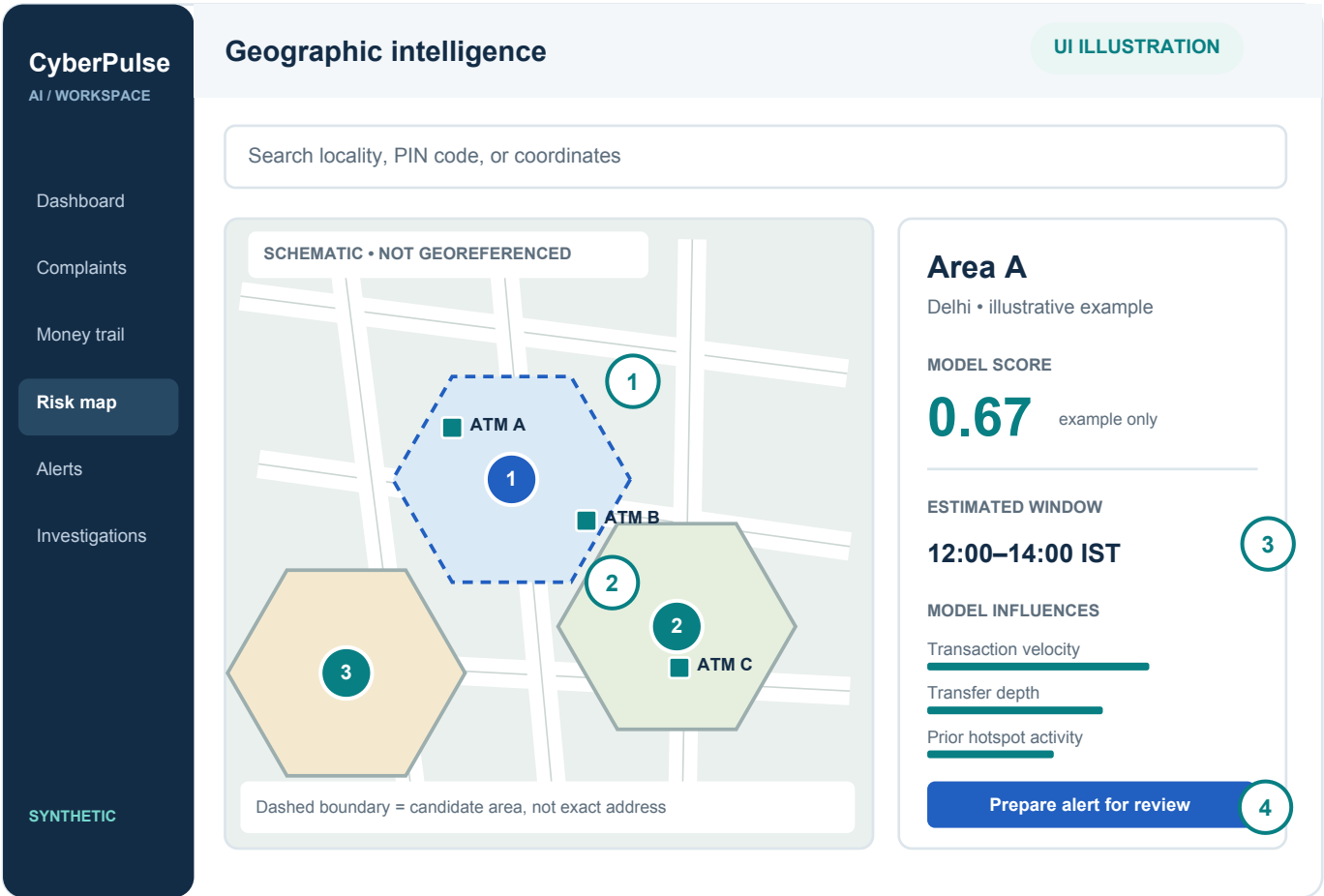


Figure 2. Annotated UI illustration, not a captured live session or an evaluated model output. ATM letters and the road layout are fictional. No map-provider data is used.

- 1

Area, not address

A candidate hexagon narrows the search. Its centre pin is a reference coordinate.
- 2

ATM and PIN context

Verify ATM records separately. A locality or PIN lookup does not locate a suspect.
- 3

Time and explanation

Show the estimated window and influential features; retain uncertainty.
- 4

Human decision

Review before action. The current prototype records alerts internally; external delivery is a pilot requirement.

03 / TECHNICAL CREDIBILITY

What exists. What needs proof.

The prototype demonstrates the workflow on synthetic data. Pilot readiness requires verified data access, stronger controls and evaluation of the full serving pipeline.

WORKING PROTOTYPE

- Complaint and money-trail views
- H3 candidate-area ranking
- XGBoost scores + temporal estimates
- SHAP feature explanations
- Internal alert records and audit events

REQUIRED BEFORE REAL-DATA USE

- Authorised bank / case-data access
- Verified identity and access isolation
- Point-in-time replay and leakage checks
- Missing-link / stale-data qualification
- Verified alert delivery and outcome capture

RECORDED SYNTHETIC BASELINE COMPARISON

Top-3 cash-out cell hit rate • model card dated 16 September 2026



The recorded XGBoost result does not exceed the simpler baselines. These numbers recover planted synthetic patterns; they are not field accuracy or proof of operational value. [4]

EVALUATE THE PIPELINE THAT OFFICERS WILL SEE

Serving combines model output with other ranking terms. Evaluate that final ranking, its candidate generator and its time window together. Enforce a prediction cutoff on historical aggregates before replay; the current hotspot query does not yet apply one. [5]

04 / EVALUATION PLAN

An eight-week gated pilot

Proposed eight-week evaluation after data-access approval. Timing, cohort size, partner ownership and numerical thresholds must be agreed before evaluation begins.

01

Readiness

Weeks 1–2

Approve data scope; verify identity, hosting, retention and prediction cutoffs.

Gate: replay-safe dataset

02

Historical replay

Weeks 3–4

Use time-ordered held-out cases. Compare the full pipeline with simple baselines.

Gate: agreed offline targets

03

Shadow testing

Weeks 5–7

Log predictions and officer review without automated operational action.

Gate: useful, timely leads

04

Joint review

Week 8

Assess benefit, alert burden and failures. Decide whether to extend or stop.

Gate: documented decision

PILOT SCORECARD

MEASURE	WHAT THE PILOT MUST ESTABLISH
Candidate recall	True cell included anywhere in the candidate set.
Top-3 / joint hit rate	Location ranking and location-plus-time success vs baseline.
Delivered lead time	Withdrawal time minus actual investigator availability time.
Investigator workload	False alerts per case/day and time spent reviewing leads.
Data / outcome coverage	Missing links, stale inputs and unobservable cash-out outcomes.

Pre-register thresholds and an alert budget; report uncertainty and all eligible cases, including candidate misses. Historical replay estimates potential lead time. Only prospective testing measures actual delivery and operational usefulness.

05 / PARTNERSHIP REQUEST

A bounded, reviewable next step

Request a technical review with RBIH and introductions to suitable bank and law-enforcement partners for a controlled evaluation. No institutional endorsement is implied.

MINIMUM DATA AND GOVERNANCE AGREEMENT

Request only the approved fields needed for evaluation: pseudonymous case/account links, transaction amounts and event/availability times, complaint timestamps, historical ATM coordinates, and linked withdrawal outcomes. Account identifiers must remain joinable under the agreed pseudonymisation scheme.

Before access: agree the institutional host, authorised users, isolation controls, audit logging, retention/deletion terms and outcome ownership. Missing outcomes remain unknown; they must not be treated as successful predictions or confirmed negatives.

SUCCESS MUST BE OBSERVABLE

Proceed only if the agreed evaluation shows useful improvement at an acceptable review burden. Higher recovery or arrest rates remain hypotheses for a later controlled operational study.

SOURCES AND IMPLEMENTATION EVIDENCE

[1] **MHA / PIB, 17 March 2026 — CFCFRMS context.**

Over Rs. 8,690 crore saved as of 31 January 2026; not a CyberPulse result.

[2] **MHA / PIB, 12 May 2026 — I4C–RBIH MoU.**

Supports mule-account intelligence sharing and fraud-risk collaboration.

[3] **RBIH — MuleHunter.ai project overview.**

Existing mule-account detection capability; referenced for operational fit.

[4] **CyberPulse model_card.json — 16 September 2026.**

apps/ml-service/models/model_card.json; synthetic data, seed 26184.

[5] **CyberPulse implementation review — 18 September 2026.**

hotspotService.ts, alertService.ts, services/lib/auth.ts and engine/temporal.py.

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